

Homework 1

Powers, factorising and algebraic fractions · about 60 minutes

Each question graded N0–E8

Answers · tutor's working, not an official mark scheme

Show all working. Leave answers with positive indices and fractions in simplest form. Each part shows the highest grade it can give: [A], [M] or [E].

Quick review

- 1 Write $\frac{1}{x^3}$ using a negative index. [A]

$$x^{-3}$$

- 2 Ana says $(a^2)^3 = a^5$. Is she right? Give a reason. [A]

No: a power of a power multiplies the indices, $(a^2)^3 = a^6$

- 3 Kai says $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$. Show why he is wrong. [A]

Lowest common denominator 6: $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$, not $\frac{2}{5}$

- 4 Which is bigger: 2^0 or 0^2 ? [A]

$2^0 = 1$ and $0^2 = 0$, so 2^0 is bigger

QUESTION ONE

- (a) Simplify each expression, leaving your answer with positive indices.

- (i) $\frac{(4a)^2}{2a^{-3}}$ [A]

$$\frac{16a^2}{2a^{-3}} = 8a^{2-(-3)} = 8a^5$$

- (ii) $\sqrt{81m^{20}n^6}$ [A]

$$\sqrt{81} = 9, \text{ halve each index: } 9m^{10}n^3$$

- (b) Factorise completely $10x^3y^2 - 25x^2y^3$. [A]

$$\text{HCF } 5x^2y^2: 5x^2y^2(2x - 5y)$$

- (c) Simplify fully, leaving your answer with positive indices. [M]

$$\left(\frac{27x^6}{x^{12}}\right)^{-\frac{1}{3}}$$

Inside the bracket first: $27x^{-6}$

$$27^{-\frac{1}{3}} \times x^{-6 \times (-\frac{1}{3})} = \frac{1}{3} \times x^2 = \frac{x^2}{3}$$

u: negative or fractional power interpreted correctly; r: correct simplified answer

- (d) Sam was asked to factorise completely (i) $2x^3 - 50x$ and (ii) $3x^2 + 5x - 12$. **Sam's answer has 2 errors.**
Find each one, fix it, and say why it loses marks. **[M]**

Sam's answer

- (i) $2x^3 - 50x = 2x(x^2 - 25) = 2x(x - 25)(x + 25)$
(ii) $3x^2 + 9x - 4x - 12 = 3x(x + 3) - 4(x - 3)$ so it won't factorise

1. (i) $x^2 - 25 = (x - 5)(x + 5)$: root the 25 too. Answer $2x(x - 5)(x + 5)$ (L2)

2. (ii) $-4x - 12 = -4(x + 3)$. Answer $(x + 3)(3x - 4)$ (L5)

- (e) Box A has sides $(x + 3)$, $(x - 2)$ and $(x + 1)$ cm. Box B has sides x , $(x + 1)$ and $(x + 2)$ cm, where $x > 2$.

- (i) Show that Box B always has the larger volume. **[E]**

$$A: (x^2 + x - 6)(x + 1) = x^3 + 2x^2 - 5x - 6; \quad B: x(x^2 + 3x + 2) = x^3 + 3x^2 + 2x$$

$$B - A = x^2 + 7x + 6. \text{ For } x > 2 \text{ every term is positive, so } B - A > 0: B \text{ is larger.}$$

- (ii) Find the value of x for which Box B's volume is 36 cm^3 more than Box A's. **[E]**

$$x^2 + 7x + 6 = 36, \quad x^2 + 7x - 30 = 0, \quad (x + 10)(x - 3) = 0$$

$$x = 3 \text{ (reject } x = -10: \text{ a side cannot be negative, and } x > 2)$$

u: (a)(i), (a)(ii), (b) each correct. r: (c) correct; (d) both errors fixed with reasons; (e)(i) both volumes expanded.
t: (e) both volumes compared with a reason, and $x = 3$ with -10 rejected. N1 partial; N2 1u; A3 2u; A4 3u; M5 1r; M6 2r; E7 t, minor error; E8 t.

QUESTION TWO

- (a) Simplify $\frac{x^2 + 4x - 21}{2x + 14}$. **[A]**

$$\frac{(x + 7)(x - 3)}{2(x + 7)} = \frac{x - 3}{2}$$

- (b) Simplify $\frac{3x^2 + 7x + 2}{9x^2 - 1}$. **[A]**

$$\frac{(3x + 1)(x + 2)}{(3x - 1)(3x + 1)} = \frac{x + 2}{3x - 1}$$

- (c) Write $\frac{2x}{x + 4} - \frac{x - 3}{x - 1}$ as a single fraction in its simplest form. **[M]**

$$\frac{2x(x - 1) - (x - 3)(x + 4)}{(x + 4)(x - 1)} = \frac{2x^2 - 2x - (x^2 + x - 12)}{(x + 4)(x - 1)}$$

$$= \frac{x^2 - 3x + 12}{(x + 4)(x - 1)}$$

u: first line set up; r: correct fraction (the numerator does not factorise)

(d) Simplify $\frac{x^2 - 9}{4x} \div \frac{x^2 + x - 6}{8x^2}$. [M]

$$\frac{(x-3)(x+3)}{4x} \times \frac{8x^2}{(x+3)(x-2)} = \frac{2x(x-3)}{x-2}$$

(e)

(i) Show that $\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$. [M]

$$\frac{1}{n} - \frac{1}{n+1} = \frac{(n+1) - n}{n(n+1)} = \frac{1}{n(n+1)}$$

(ii) Hence find the exact value of $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{9 \times 10}$. [E]

$$\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{9} - \frac{1}{10}\right)$$

$$= \frac{1}{2} - \frac{1}{10} = \frac{2}{5}$$

Every middle term cancels with the next one; only the first and last are left.

u: (a), (b) correct; (c) first line set up. r: (c) correct; (d) correct; (e)(i) shown.

t: (e)(ii) $\frac{2}{5}$ with the cancelling explained. N1 partial; N2 1u; A3 2u; A4 3u; M5 1r; M6 2r; E7 t, minor error; E8 t.

QUESTION THREE

(a) Solve $\frac{4x+3}{x-5} - 2 = 0$. [A]

$$4x + 3 - 2(x - 5) = 0, \quad 2x + 13 = 0$$

$$x = -\frac{13}{2}$$

(b) Solve $\frac{x+2}{3} - \frac{x-1}{4} = 2$. [A]

$$\text{Multiply every term by 12: } 4(x+2) - 3(x-1) = 24, \quad x+11 = 24, \quad x = 13$$

(c) Make r the subject of $P = \frac{2r+5}{r-3}$. [M]

$$Pr - 3P = 2r + 5, \quad Pr - 2r = 3P + 5, \quad r(P-2) = 3P + 5$$

$$r = \frac{3P+5}{P-2}$$

(d) Solve $\frac{9}{x} - \frac{4}{x+1} = 2$. [M]

Multiply every term by $x(x+1)$: $9(x+1) - 4x = 2x(x+1)$

$$5x + 9 = 2x^2 + 2x, \quad 2x^2 - 3x - 9 = 0, \quad (2x+3)(x-3) = 0$$

$$x = 3 \quad \text{or} \quad x = -\frac{3}{2}$$

(e) A kayaker paddles 16 km up a river against a current of 2 km/h, then 16 km back down with the current. The whole trip takes 6 hours. Her speed in still water is v km/h.

(i) Show that $3v^2 - 16v - 12 = 0$. [M]

$$\frac{16}{v-2} + \frac{16}{v+2} = 6$$

$$16(v+2) + 16(v-2) = 6(v-2)(v+2), \quad 32v = 6v^2 - 24, \quad 3v^2 - 16v - 12 = 0$$

(ii) Find her speed in still water. Explain why there is only one answer. [E]

$$(3v+2)(v-6) = 0, \text{ so } v = 6 \text{ or } v = -\frac{2}{3}$$

A speed cannot be negative (and she must beat the current, $v > 2$), so $v = 6$ km/h.

u: (a), (b) correct; (c) first step. r: (c) correct; (d) both solutions; (e)(i) derived clearly.

t: (e)(ii) $v = 6$ with the negative root rejected for a reason. N1 partial; N2 1u; A3 2u; A4 3u; M5 1r; M6 2r; E7 t, minor error; E8 t.

Totals out of 24. Cut scores move a little each year: Achievement from about 7–8, Merit 13–14, Excellence 19–20.